



Measuring the speed of sound in propane gas using a Rubens tube

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In this report, the speed of sound in a Rubens tube is determined by varying the frequency of a tone applied to the exterior of one of the tube's ends and measuring the wavelength of the standing waves produced by the release of the flammable gas through a line of holes at the top of the tube. We use the Phyphox application to determine the tone frequency causing standing waves across the tube and make approximate measurements of the wavelength at these frequencies with a meter stick. The wave relationship tells us that only the wavelength and frequency of a wave are required to determine its speed so, with this data, we are able to determine the speed of sound in the propane gas inside the tube. Through our method, we find this wave speed to be around 210 ± 5 m/s, which is 18% smaller than the currently accepted value.

I. INTRODUCTION

The Rubens tube offers a unique and engaging opportunity to indirectly measure the speed of sound in propane in a relatively simple lab setting. Measuring the speed of sound in a material can often be a difficult procedure but the design of the Rubens tube allows one to determine an approximate value for the sound wave speed inside flammable gases. The Rubens tube utilizes the standing pressure waves, waves whose amplitude profile does not move in space, created inside of a cylindrical tube from an externally applied source of noise by filling it with flammable gas. The gas is allowed to escape the tube through a line of small holes drilled into the top of the tube. These streams of gas are then lit on fire to visualize how the wave patterns change within the tube as the noise source changes in pitch.

In 1905, Heinrich Rubens built the first Rubens tube, which he called a “flame tube” to visually demonstrate standing waves for his undergraduate students [1]. The tube was designed to be a simple demonstration of standing waves that could allow one to make measurements of the frequency and wavelength of these waves. Though the tube features relatively simple wave phenomena, it should be noted that the dynamics of the flames from the Rubens tube are currently not well-understood [2].

In this report, we exploit the Rubens tube's ability to visually display standing waves that occur inside the cylindrical tube to measure the speed of sound inside of it. We vary the frequency being applied to the tube and record the wavelength of the standing waves of fire coming from the tube. With this data, we can estimate the speed of sound, c , inside this cylinder using the basic traveling wave relations.

II. ANALYSIS

Given the standing wave relationship

$$c = \lambda f, \quad (1)$$

if one makes measurements of a wave's frequency f and its wavelength λ , the wave's speed of propagation in a given medium, c , can be determined. It should also be noted that the inverse of a wave's frequency is its period:

$$T \equiv \frac{1}{f} \quad (2)$$

since the frequency of a wave captures its number of cycles per second and the period captures the number of seconds per cycle.

In this work, we attached a signal generator at one end of the cylindrical tube that allowed us to vary the frequency of the tone being applied. The frequency was determined using a sensor application called Phyphox and, since the tone being emitted was pure, we assumed that there would be no uncertainty in this measurement. When the applied tone resulted in standing pressure waves inside of the tube, visually determined by the standing waves of fire coming from the top of the tube, the wavelength was manually determined by two methods. We first counted the number of flame peaks across the Rubens tube and divided the total length of the cylinder by this number. Our second method involved directly measuring the wavelength from one fire node to another. Since the assumption that *all* of the waves would be visible across the tube; that there wouldn't be parts of the waves cut off near the source of noise or near the end of the tube, and the assumption that the direct measurement of wavelengths in our second method were exact cannot be confidently made, we have, somewhat arbitrarily, assumed an error in the determined wavelength from both of these methods to be around 3 centimeters.

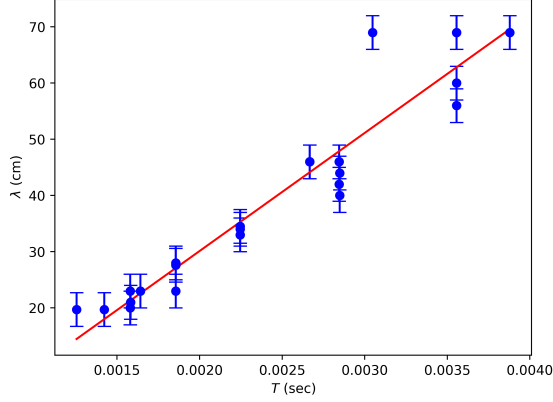


FIG. 1. Our collected period and wavelength data (wavelength data is coming from both our measurement methods) with a linear fit. The slope of this line contains our measurement of the speed of sound in propane gas. We find that $c \approx 210 \pm 5$ m/s. This line of best fit is found to have a reduced chi-squared value of .46.

With a range of measured frequency and wavelength values, if we wish to infer the speed of sound inside of this tube from the collected data, we can plot the inverse of the frequency (the period of the wave, see Eq. (2)) by the wavelength of each observation and fit a polynomial defined by $\lambda = c/f \equiv cT$ to this data. The slope of this function fit to our T and λ data will give us an inferred speed of sound, c , inside the tube. This choice of axes is made because we have assumed zero error in the frequency measurements and we have uniform uncertainties in the wavelength measurements. Combining all of the collected wavelength and period data collected in this experiment, the following plot is made: From the slope of this plot, we are able to determine the speed of sound in propane gas. We find that this speed c is around 210 ± 5 m/s. The speed of sound in propane is currently accepted to be 258 m/s [3], which means that our measured value is around 18% smaller than expected. From this plot, we also compute a reduced chi-squared for the linear line's fit to the data since it allows one to quantitatively determine how well-aligned this fit is to the data and because our data has considerable error bars. The computation of the reduced chi-squared is demonstrated in Appendix A. We find that the reduced chi-squared value is .46, meaning that this linear line makes a reasonable fit to the collected data, given the uncertainty in the wavelength measurements, since a reduced chi-squared value of 0 would indicate a perfect fit to the data while values greater than two would mean that the data falls more than two standard deviations away from the best-fit line, which would be an inaccurate fit. If the wavelength values were more accurately measured, it would be expected that this fit would become more aligned with the data, which would result in a higher reduced chi-squared value.

III. CONCLUSIONS

In this work, the speed of sound inside of a Rubens tube is determined by repeated measurements of its standing waves' wavelength resulting from an applied source of noise with a pure frequency. Given the standing wave relationship, the speed of a wave can be determined by just measuring its wavelength and frequency. The Rubens offers a simple experimental design to roughly measure these properties of sound waves in propane since this is the gas required for the Rubens tube. We find the speed of sound in propane to be 210 m/s, which is 18% smaller than expected.

The result of this study, although approximate, is significant because it shows that one can determine a complicated property of a flammable gas, such as propane, through basic measurements of a Rubens tube. To make a more precise measurement of the sound wave speed in propane with the Rubens tube, a more accurate method of determining the wavelength of the sound wave inside of the wave will need to be implemented. Our measurements of the wavelength were very approximate; determined through the use of a meter stick. Another avenue for making a more precise measurement of this wave speed in the Rubens tube may include designing a longer tube and one that has a higher pressure of gas so that the nodes and peaks of the standing waves are made more apparent.

Appendix A: Uncertainty Analysis

The general formula for the uncertainty in a function with n variables, $f(x^0, x^1, \dots, x^n)$, which all have uncertainties, $\delta x^0, \delta x^1, \dots, \delta x^n$, is given as [4]:

$$\delta f = \sqrt{\sum_{i=0}^n \left(\frac{\partial f}{\partial x^i} \delta x^i \right)^2} \quad (\text{A1})$$

Since we have assumed that there is no uncertainty in frequency, the uncertainty in the speed of light will be of the form:

$$\delta c = \frac{\partial c}{\partial \lambda} \delta \lambda = f \delta \lambda \quad (\text{A2})$$

Since we have multiple measurements of f , the uncertainty in c is determined by summing over each f_i value and dividing by the total number of measurements, N :

$$\delta c = \frac{\sum_{i=0}^N f_i \delta \lambda}{N} \quad (\text{A3})$$

The reduced chi-squared is computed by:

$$\chi^2_\nu = \frac{\chi^2}{N - P} \quad (\text{A4})$$

where N is the number of measurements made, P is the number of parameters used to construct a fitting model

(we use $P = 2$ for the slope and y-intercept of the linear function shown in Fig 1), and χ^2 is the regular chi-squared, defined as:

$$\chi^2 = \sum_{i=0}^N \left(\frac{y_i - f_i}{d_i} \right)^2 \quad (\text{A5})$$

where y_i represents the observed values of the parameter you're attempting to fit to, f_i is the value of the function you're trying to fit the data to (at the same index of the y_i -value), and d_i is the uncertainty for the observed

values at index i .

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